Niranjan Madhavan Dindukurthu

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SUMMARY

Site Reliability Engineer / Production Engineer with 5+ years of experience improving the reliability, scalability, and operational excellence of large-scale distributed systems, focused on Ads and infrastructure at Meta, Google, and Cisco.

EXPERIENCE

Meta - SRE / Production Engineer London, UK
Ads - Catalog June '25 - present

- Prevented high-severity production incidents by building automated regression detection and validation gates that replay production traffic against code changes, blocking broken changes before they reach production.
- Reduced incident detection time by 40% by engineering automated rollout controls with real-time SLI monitoring and automated rollback on metric regression, enforcing error budget policies and operational readiness gates across every feature release at Meta.
- Cut $0.5M/yr in compute by building an AI-powered performance optimization system that leveraged profiled production workloads to identify bottlenecks and hot code paths, automatically generating optimized code changes across services.
- Built end-to-end test automation frameworks integrated into release pipelines, increasing test coverage from <5% to 95% across critical ad-serving services. Cut QA costs by ~$150K/year while reducing incident detection time from weeks to minutes.
- Drove 25% reduction in mean-time-to-detection and 30% faster resolution by building unified observability (SLIs/SLOs, metrics, alerts, dashboards) and end-to-end distributed tracing, enabling faster root cause analysis across ad-serving services.
- Served on the on-call rotation for critical ad-serving services, leading incident response during high-severity outages and driving blameless root cause analysis and corrective actions that reduced repeat incidents across teams.

Google - Systems Development Engineer Bangalore, India
GCP - Google Distributed Cloud Hosted May '24 - May '25

- Accelerated client deployments by 30% and reduced failures by 40% by engineering a metric-driven hardware validation system for Google's private cloud, integrated into the deployment pipeline.
- Designed and deployed a fleet-wide infrastructure observability platform using custom exporters across diverse hardware classes, standardising ingestion into a unified monitoring and alerting system.
- Performed capacity planning and load analysis for GDCH deployments, modelling resource utilisation trends to right-size infrastructure and prevent capacity-related outages before customer impact.

Cisco - SRE Intern → SRE I → SRE II India (Remote)
Network Engineering and Operations Feb '21 - Mar '24

- Modernized core network infrastructure by migrating legacy IPAM systems to a scalable platform with an event-driven DNS backend processing millions of events per day, improving scalability and reliability for Cisco's global network.
- Managed on-prem Kubernetes clusters across multiple regions running production network services, maintaining cluster health and resolving pod/node-level failures to ensure high service availability.
- Eliminated cascading failures and improved reliability of the DNS service handling 2M QPS, by implementing client attributions and rate limiting on the provisioning control plane, reducing service disruptions under high load.
- Cut MTTR by 60% by automating incident resolution workflows and building operational runbooks that enabled junior engineers to resolve incidents independently.

Manipal Institute of Technology - Research Assistant Manipal, India
Software Defined Networks (SDN) - Security Jan '20 - Jul '21

- Built a statistical intrusion detection system for SDN environments using DPDK-enabled Open vSwitch, detecting DDoS attacks 35% faster than traditional detection systems.

AWARDS AND RECOGNITION

- Received Amaze 2 award for **'Driving the highest level of achievement'** for driving the IPAM integration project at Cisco.

SKILLS

- **Languages:** Python, Go, Bash, SQL, Typescript.
- **Systems:** Distributed systems design, fault-tolerant architectures, capacity planning, SLI/SLO frameworks.
- **Infra/Platform:** Linux/Unix, Kubernetes, Docker, GCP, Terraform, Ansible, Helm, Jenkins, CI/CD, Git.
- **Observability:** Prometheus, Grafana, Splunk, distributed tracing, alerting, SLO-based monitoring.

CERTIFICATIONS

- Cisco Certified Devnet Associate

EDUCATION

Manipal Institute of Technology, MAHE Manipal, India

- Bachelor of Technology in Computer and Communication Engineering Aug '21
- Coursework**:** Distributed Systems, Data Structures and Algorithms, Computer Networks, Operating Systems, Database Management.